SDC

Hackathon

Technical Tools Declaration

1.0 TEAM AND PROJECT OVERVIEW

Team Name: kpmbians

Selected Domain: Domain 2 – Orbyte Smart SME Challenge

Selected SDG Track (1–17): SDG 8 – Decent Work and Economic Growth

Project / Solution Name: TwinBoss

Name (Leader): NUR ALIA MAISARAH BINTI MUHAMMAD AMIN

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University Affiliation(s): Kolej Profesional MARA Beranang

Number of Team Members: 2

Team Members:

1. Nur Alia Maisarah binti Muhammad Amin (Team Leader)
2. Nur Husnina Hidayah binti Abdul Razi

Live Prototype URL: https://twin-boss-mvp.replit.app

Code Repository URL: https://github.com/nurhusninahidayah-dev/TwinBoss_MVP.git

2.0 ARCHITECTURE & TECHNOLOGY STACK

2.1 System Architecture Overview

TwinBoss is a web-based application designed to streamline SME refund management through AI-assisted assessment and a structured approval workflow. The frontend is developed using React 18, Vite, TypeScript, and Tailwind CSS, providing role-specific interfaces for customers, staff, and business owners.

When a customer submits a refund request, the application processes the submitted information and retrieves relevant order and customer details from locally defined mock data. The request information is then sent to the Groq API, which provides access to the openai/gpt-oss-20b model for AI-assisted reasoning and refund assessment.

The AI-generated assessment supports staff verification and provides recommendations for the business owner's review. Once an authorized decision is made, staff can proceed with the refund workflow and customer communication.

The prototype uses simulated business records rather than a live Orbyte database integration. Application state is managed using React's useState, and the development and deployment environment is provided through Replit, subject to the actual deployment configuration.

System Flow: User Interface → Refund Processing Logic → Mock Business Data and Groq API → AI Assessment → Staff Verification → Owner Approval → Staff Execution.

2.2 Languages, Frameworks & Libraries

Name	Purpose in Project	License
React 18	Builds the user interface and role-specific application views.	MIT
Vite	Supports frontend development and application bundling.	MIT
TypeScript	Implements application logic and provides static type checking.	Apache 2.0
Tailwind CSS	Provides styling and responsive interface layouts.	MIT
Lucide Icons	Provides icons for interface elements and visual indicators.	ISC

3.0 APIs & CLOUD SERVICES

3.1 External APIs

TwinBoss uses the Groq API to access a large language model for AI-assisted refund assessment, risk identification, and recommendation generation. The API processes relevant information from refund requests and returns AI-generated assessments to support staff verification and business owner approval.

API and Provider	Access Tier (Free / Paid / Sponsor-Provided)	Purpose in Project
Groq API — Groq, Inc.	Free tier	Provides access to the openai/gpt-oss-20b model for AI-assisted reasoning, refund assessment, and text generation.

API Security: API keys and other credentials are excluded from this declaration and should be stored securely rather than exposed in a public code repository.

3.2 Cloud Platforms & Deployment

Replit was used as the development environment for the TwinBoss prototype. Replit Agent assisted with code generation, while the team reviewed and modified the generated code during development. Replit was also used to deploy the web application as a static website for demonstration purposes.

The prototype uses synthetic mock business data to simulate order and customer records. It does not integrate directly with a live Orbyte database.

Service and Provider	Purpose in Project	Plan / Region
Replit	AI-assisted code generation, application development, testing, and static website deployment.	Replit Core (provided free through SDC); United States (default Replit Cloud region).

The live prototype URL will be provided in Section 1. The application should remain accessible to the judges during evaluation.

4.0 AI TOOLS, MODELS & DATASETS

4.1 AI Models Used in the Solution

TwinBoss uses a pre-trained large language model accessed through the Groq API to support AI-assisted refund assessment, risk identification, and recommendation generation. The model processes relevant information from refund requests and generates responses to assist staff verification and business owner decision-making.

Model & Version	**Source / Provider**	**License / Terms**	**Usage (Pre-trained / Fine-tuned / Self-trained)**
openai/gpt-oss-20b	model accessed through the Groq API	Apache License 2.0; Groq API usage subject to the provider's applicable terms	Pre-trained

4.2 Datasets

TwinBoss does not use public or private real-world datasets for training, fine-tuning, evaluation, or application operation. Instead, the team created synthetic mock business data to simulate customer information, order records, and refund scenarios for prototype demonstration and testing.

The synthetic data does not originate from actual customers or businesses and is intended solely to demonstrate the application's workflow.

Dataset Name	Source / URL	Type	License	Personal Data? (Y/N)
TwinBoss Synthetic Mock Business Data	Created by the team; no external source	Synthetic	Team-created	No

4.3 AI-Assisted Development Tools

Did your team use any generative-AI tool to help create code, design, text, or other project material?

Answer: Yes

Tool	Extent of Use (Minor / Moderate / Major)	Used For
Replit Agent	Major	Assisted with generating initial application code and implementing features for the TwinBoss prototype. The team reviewed, edited, and refined the generated code to meet project requirements.
DeepSeek	Moderate	Used to assist with coding tasks, explore implementation approaches, and support debugging during the development of the TwinBoss application.

ChatGPT	Moderate	Used to assist with coding-related questions, explain technical concepts, suggest implementation approaches, and support the preparation and refinement of project documentation.

Review, Testing, and Responsibility for AI-Generated Output

The team reviewed and modified AI-generated code during development to align it with the intended system requirements and workflow. The application was tested through prototype development and functional checks to identify issues and verify expected behaviour. The team remains responsible for the accuracy, functionality, and quality of the submitted solution, including the code generated with AI assistance.

5.0 SPONSOR PROVIDED RESOURCES

5.1 Did your project use any official SDC sponsor-provided resource?

Yes. The TwinBoss project used Replit Core access provided free of charge through SDC. Replit served as the team's primary development environment throughout the development of the prototype. The team used Replit Agent to assist with generating initial application code, while team members reviewed, edited, and refined the generated code to meet the project's requirements. Replit was also used to develop, test, and deploy the TwinBoss web application as a static website. This resource helped streamline the development process and enabled the team to implement and demonstrate the proposed AI-powered refund management solution within the hackathon timeframe.

5.2 Resources Used

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
SDC	Replit Core access provided free of charge through SDC	Replit Core was used as the team's development platform for building the TwinBoss web application. Through Replit, the team used Replit Agent to assist with generating initial code and accelerating the implementation of application features. The team subsequently reviewed, edited, and refined the code to align with the intended system workflow and requirements. Replit was also used to support application development, testing, and static website

		deployment, allowing the team to make the TwinBoss prototype accessible for demonstration.

Supporting Evidence: Replit Development Environment

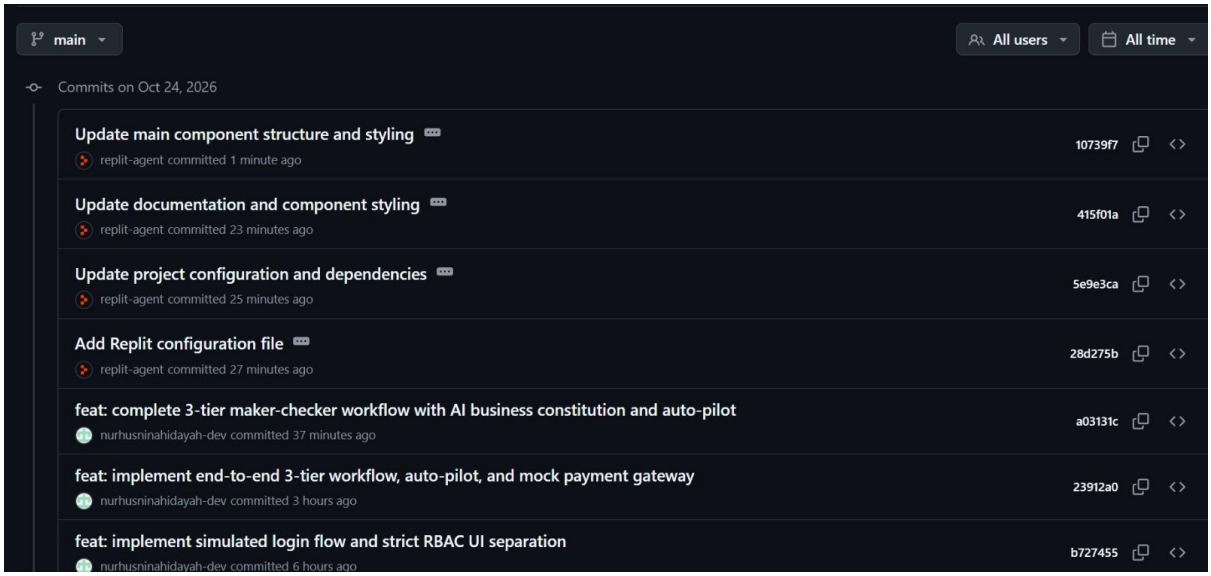


Figure 5.1: Replit development environment used by the team to develop, refine, and deploy the TwinBoss prototype.

5.3 Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced?

Replit Core was a valuable resource that supported the development and deployment of the TwinBoss prototype. Its integrated development environment and Replit Agent helped the team generate initial code, implement application features, and refine the system more efficiently. However, Replit Core was not essential to the core functionality of TwinBoss itself, as the proposed refund management workflow and AI-assisted assessment could be implemented using alternative development tools and hosting platforms. Therefore, Replit Core primarily served as a development and deployment resource that accelerated implementation rather than a mandatory component of the solution's underlying functionality.

6.0 SECURITY, PRIVACY & RESPONSIBLE USE

6.1 Data Handling & Privacy

TwinBoss currently uses synthetic mock business data created by the team for demonstration and testing purposes. The prototype does not rely on real customer records, actual financial information, or a live Orbyte database. The mock data simulates order details and customer information to demonstrate the refund assessment workflow.

The application uses the Groq API to access the openai/gpt-oss-20b model for AI-assisted refund assessment. Information submitted for AI assessment should therefore be limited to the synthetic data used in the prototype. No real personal or sensitive customer information should be entered during demonstrations or testing.

As the current prototype is intended for hackathon demonstration rather than production use, it should not be used to process actual customer refunds or confidential business information. Before production deployment, additional safeguards would be required, including secure API key management, appropriate access controls, data protection measures, and a review of the external API provider's data-handling terms.

6.2 Known Limitations & Risks

TwinBoss is a functional prototype with several limitations. First, its business records are simulated using synthetic mock data, so the assessment results do not demonstrate integration with a live Orbyte database or actual business records. Second, AI-generated assessments may be inaccurate, inconsistent, or influenced by incomplete information. The AI assessment should therefore be treated as decision-support information rather than definitive proof that a refund request is valid or fraudulent.

The prototype also depends on the availability of the Groq API and may be affected by network interruptions, API rate limits, or service availability. Refund processing and payment-related activities are demonstrated through simulated workflows rather than

a fully integrated payment gateway. Similarly, customer communication features should not be assumed to deliver messages automatically unless the relevant integration is implemented.

Human verification and business-owner approval remain important safeguards. Before production use, the system would require more extensive testing, security reviews, reliable backend handling of API credentials, and integration testing with actual business systems.

7.0 COMPLIANCE & READINESS CHECKLIST

Please confirm that the following requirements have been fulfilled before submitting the Technical Tools Declaration.

☒ All external APIs and cloud services used in the project have been declared.

☒ All AI models, datasets, and AI-assisted development tools have been declared, including Replit Agent, DeepSeek, and GPT.

☒ Third-party libraries and tools have been identified with their applicable licences or terms of use.

☒ API keys, passwords, and other confidential credentials have not been included in this declaration.

☒ AI-assisted development has been disclosed, including the use of generative-AI tools to assist with code generation, technical problem-solving, and project documentation.

☒ All official SDC sponsor-provided resources used by the team have been declared, including Replit Core access provided through SDC.

☒ The prototype and code repository links have been provided and checked for accessibility by the judges.

☒ All sections of this declaration have been completed accurately, and the information is consistent with the actual implementation of TwinBoss.

8.0 DECLARATION

I hereby declare that the information provided in this Technical Tools Declaration is accurate and complete to the best of my knowledge. I acknowledge that all external APIs, cloud services, AI models, datasets, third-party libraries, AI-assisted development tools, and sponsor-provided resources used in the TwinBoss project must be disclosed appropriately. I understand that the team is responsible for ensuring that the prototype complies with applicable licensing requirements, security considerations, and the competition's rules.

Team Leader: NUR ALIA MAISARAH BINTI MUHAMMAD AMIN

Project Name: TwinBoss

Team Name: kpmbians

Signature: 

Date: 24/10/2026